

Module II – Programming Labs (Python & R)

- Industry-Style Coding Assignments

LAB 1: Python & R Basics

- Objective: Learn basic syntax and data handling
- Dataset: Customer_Analytics_Module_II.csv
- Focus on variables, data types, and commands

LAB 1: Tasks

- Load dataset in Python or R
- Display first 10 rows
- Calculate average age and income
- Count churn vs non-churn customers

LAB 1: Deliverables

- Working script or notebook
- Output screenshots
- Short explanation of results

LAB 2: Data Manipulation & Visualization

- Objective: Perform industry-style data manipulation
- Focus on Pandas / dplyr
- Business interpretation required

LAB 2: Tasks

- Filter high-income customers
- Group by gender and compute averages
- Create bar and scatter plots
- Interpret churn patterns

LAB 2: Deliverables

- Clean, well-commented code
- Minimum 2 plots
- Business insight summary

LAB 3 : Reproducible Analytics

- Objective: Analyze reproducibility and performance
- Compare Python vs R workflows
- Critical evaluation required

LAB 3: Tasks

- Benchmark execution time
- Analyze memory usage
- Discuss reproducibility challenges
- Propose best practices

LAB 3 : Deliverables

- Research-style report
- Performance comparison table
- Critical discussion